

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 415 — Past Questions & Materials
by Renance

Course: MTH 415
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

This study pack was compiled and restructured by Renance for MTH 415 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — MTH 415 Fluid Dynamics 1 2020 2021 First Semester

Level 400 • First Semester • PastQuestion

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DERAL UNIVERSITY of PETROLEUM RESOURCES, EFFURUN, NIGERIA

COLLEGE OF SCIENCE

- DEPARTMENT OF MATHEMATICS

C (MATHEMATICS) DEGREE EXAMINATION (2020/2021), FIRST SEMESTER.

MTH 415 (Fluid Dynamics | 3C). November, 2021. Duration: 2hrs

Instructions: Answer ANY FOUR (4) questi

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All questions Carry equal Marks

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motion. These are the: 24> j3 (s)

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Eulerian framework: In this approach, the quis considered as a whole. Global vector and scalar variables such as velocity and temperature are defined and solved for over the entire flow-field simultaneously. This is by far the most widely used. 4

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, individual fluid particles are tracked. Each particle is

(b) Sate the Bernoulli's theorem ina Steady, inviscid flow with no body forces.

Solution: Bernoulli's theorem states that in a Steady, inviscid flow with no body forces acting, the pressure is related to the velocity via $P_v -$

1 (3/

$p + \frac{1}{2} \rho v^2 = \text{constant along a coum } <i$

(c) By definition, $c^* = m$, obtain an expression for Mach number.

Solution: By definition, 2, \

dp

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$m_5 \cdot \backslash We$

where c is the speed of sound. Thus a small change in density is related to a small change in pressure via

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and so | f

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- = OC) = O(M), MA

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on defining the Mach number

Paper 2 of 2 — MTH 415/413 Complex Analysis 11 2018/2019 2020/2021 First Semester

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2018/2019

Course code & title: MTH 415/MTH 313 (Complex Analysis II) Duration: 2 ½ hrs

Instruction: Answer any 4 (four) questions only

Question One

- a. Prove the fundamental theorem of algebra: Every polynomial equation $P(z) = a_0 + a_1z + a_2z^2 + \dots + a_nz^n = 0$ where the degree $n \geq 1$ and $a_n \neq 0$ has at least one root
 Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series valid for:
 - b. $1 < |z| < 3$
 - c. $|z| > 3$

Question Two

Let $w = f(z) = (z^2 + 1)^{\frac{1}{2}}$.

- a. Show that $z = \pm i$ are branch points of $f(z)$ and that a complete circuit around both branch points produces no change in the branches of $f(z)$
- b. Determine the branch lines for the function $f(z)$
- c. Discuss the Riemann surface for the function $f(z)$

Question Three

- a. Prove Rouché's theorem: Suppose $f(z)$ and $g(z)$ are analytic inside and on a simple closed curve C and suppose $|g(z)| < |f(z)|$ on C . Then $f(z) + g(z)$ and $f(z)$ have the same number of zeros inside C .
 Find the residues of
 - b. $f(z) = \frac{z^2 - 2z}{(z+1)^2(z^2+4)}$
 - c. $f(z) = e^z \csc^2 z$ at all its poles in the finite plane

Question Four

- a. Find the Laurent series about the indicated singularity of the function $\frac{1}{z^2(z-3)^2}$; $z = 3$ and $z = 0$.
- b. Name the singularity in each case in (a) and give the region of convergence of each series.
- c. Evaluate the integral $\int_0^\infty \frac{dx}{x^6 + 1}$

Question Five

- a. Prove the Argument theorem: Let $f(z)$ be analytic inside and on a simple closed curve C except for a pole $z = \alpha$ of order (multiplicity) p inside C . Suppose also that inside C , $f(z)$ has only one zero $z = \beta$ of order (multiplicity) n and no zeros on C . Prove that $\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)} dz = n - p$
- b. Prove that the series $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$
- c. Evaluate $\int_0^{2\pi} \frac{d\theta}{3 - 2\cos\theta + \sin\theta}$

Question Six

- a. Find the Laurent series about the indicated singularity of the function $(z-3)\sin\frac{1}{z+2}$; $z = -2$.
- b. Name the singularity in each case in (a) and give the region of convergence of each series.
- c. Prove that the series $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots = \frac{\pi^3}{32}$